

Question	Answer	Marks	Guidance
1(a)	product of mass and velocity	B1	not speed for velocity
1(b)(i)	$a = \Delta v / (\Delta t)$ or $= (v - u) / (\Delta t)$ or = gradient of slope	C1	
	acceleration = $(-0.8 - 1.6) / (0.12 - 0.09)$ = $(-)80 \text{ m s}^{-2}$	A1	No tolerance.
1(b)(ii)	0.15×1.6 or 0.15×0.8	C1	No tolerance. ECF from (b)(i) for v read-offs
	change in momentum = $0.15 \times (-0.8 - 1.6)$ = $(-)0.36 \text{ kg m s}^{-1}$	A1	Ignore sign
	OR $F = 0.15 \times (-)80$ = $(-)12$	(C1)	ECF a from (b)(i)
	Change in momentum = $(-)12 \times (0.12 - 0.09)$ = $(-)0.36 \text{ kg m s}^{-1}$	(A1)	
1(b)(iii)	$\frac{1}{2} \times 1.60 \times 0.020$ or $1.6^2 / (2 \times 80)$ or $1.6 \times 0.02 - \frac{1}{2} \times 80 \times 0.02^2$ or $\frac{1}{2} \times 80 \times 0.02^2$ = 0.016	C1	ECF from (b)(i) for v read-offs ECF a from (b)(i) .
	OR $\frac{1}{2} \times 0.80 \times 0.010$ or $0.8^2 / (2 \times 8)$ or $-0.8 \times 0.01 + \frac{1}{2} \times 80 \times 0.01^2$ or $\frac{1}{2} \times 80 \times 0.01^2$ = 0.004	(C1)	
	distance = $0.016 + 0.004$ = 0.020 m	A1	Allow 0.02 (1SF) treat finding displacement as XP e.g. $(-0.8)^2 - 1.6^2 / 2 \times 80 = 0.12 \text{ m}$

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1(c)(i)	relative speed of approach = relative speed of separation	B1	Allow 'relative speed before collision = relative speed after collision' Allow $u_1 - u_2 = v_2 - v_1$ not relative <u>velocity</u> for speed, but allow relative velocity of approach = – relative velocity of separation
	(speed of separation = 1.6 m s^{-1} so) $v_Y - (-0.80) = 1.6$, so $v_Y = 0.80 \text{ m s}^{-1}$	B1	Allow $v_Y + 0.80 = 1.6$ Allow $0.80 + 0.80 = 1.6$ BOD Ignore $v_Y = 1.6 / 2$
1(c)(ii)	a straight horizontal line from (0, 0) to (0.09, 0) or a straight horizontal line from (0.12, 0.8) to (0.20, 0.8)	B1	tolerance $\pm \frac{1}{2}$ small square
	Both horizontal lines correct <u>and</u> straight diagonal line from (0.09, 0) to (0.12, 0.8)	B1	Allow horizontal line from (0, 0) to (0.09, 0) to be missing if diagonal line starts at (0.09, 0)
1(c)(iii)	$(0.15 \times 1.6) = (0.15 \times (-0.80)) + 0.80m$ $0.36 = 0.8m$	C1	ECF from (b)(i) for v read-offs
	OR $(\frac{1}{2} \times 0.15 \times 1.6^2) = (\frac{1}{2} \times 0.15 \times 0.80^2) + (\frac{1}{2} \times m \times 0.80^2)$ $0.192 = 0.048 + 0.32m$	(C1)	Allow with the $\frac{1}{2}$ factor missing from all three terms.
	OR $m = 0.15 \times 80 / [0.80 / (0.12 - 0.09)]$ $m = 12 / 26.7$	(C1)	
	mass = 0.45 kg	A1	